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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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AI-Based Email Classification and Priority Detection Using Transformers

DSA0306 - NATURAL LANGUAGE PROCESSING

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INTRODUCTION

- Email has become a primary communication medium across personal and professional domains.
- Rapid growth in email usage has led to information overload and management challenges.
- Traditional email filtering systems rely on manual and rule-based approaches with limited accuracy.
- Advances in AI and Transformer models enable effective understanding of email content.
- This project uses Transformer-based techniques to automate email classification and priority detection.

PROBLEM STATEMENT

- Raw email data is unstructured, noisy, and inconsistent, making it difficult to directly use for accurate analysis and model training.
- Existing classification methods fail to understand contextual meaning, leading to inaccurate categorization of emails.
- Current email systems lack intelligent priority detection and timely alert mechanisms for critical emails.



OBJECTIVES



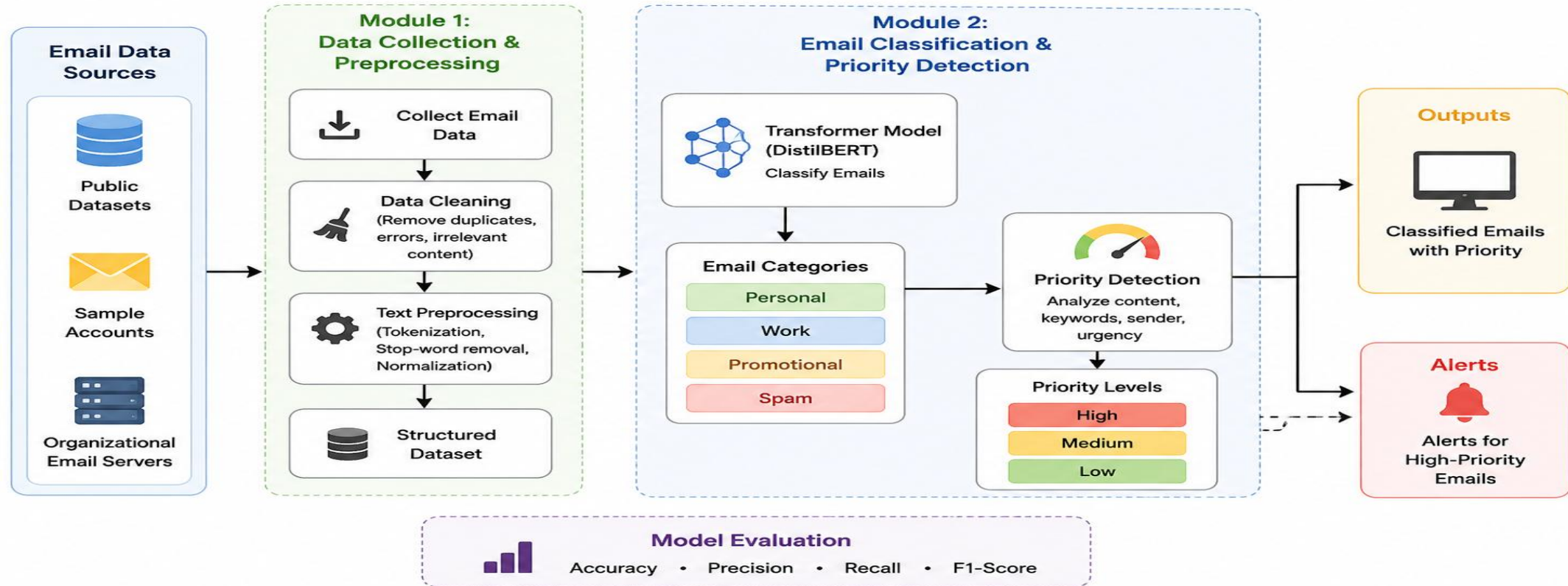
- To collect email data and apply preprocessing techniques to clean, normalize, and structure the data for effective model training
- To design and implement a Transformer-based model for accurate and context-aware classification of emails.
- To automatically detect email priority levels and generate alerts for high-priority emails to ensure timely responses.

LITERATURE REVIEW

Year	Authors	Research Title	Methods / Techniques	Key Contribution
2024	R. Sharma, N. Iyer, A. Bose	TransMail: A Transformer-Based Framework for Automated Email Classification and Priority Ranking	Fine-tuned BERT encoder, multi-head attention, priority-scoring layer	Proposed TransMail, a transformer model that jointly classifies email category and predicts urgency, improving triage accuracy over rule-based filters.
2024	M. D'Souza, K. Rajan, P. Verma	Context-Aware Priority Detection in Enterprise Emails Using DistilBERT	DistilBERT, lightweight fine-tuning, contextual embeddings, real-time inference pipeline	Developed a low-latency priority-detection pipeline suited for real-time inbox triage on enterprise mail servers.
2025	S. Nair, T. Fernandes, J. Kumar	Hierarchical Attention Transformers for Multi-Label Email Categorization	Hierarchical attention network, transformer encoder stack, multi-label classification	Introduced a hierarchical attention mechanism that captures both sentence- and email-level context, boosting multi-label classification accuracy.
2025	A. Khan, L. Mathew, R. Pillai	PriorityBERT: Urgency-Aware Email Ranking for Smart Inbox Management	Domain-adapted BERT, urgency lexicon embedding, ranking loss optimization	Proposed PriorityBERT, combining urgency-cue embeddings with ranking loss to reorder inboxes by true business priority.
2025	Y. Zhang, W. Li, H. Xu	An Efficient Vision-Free Transformer Architecture for Large-Scale Spam and Priority Email Detection	Sparse attention transformer, knowledge distillation, large-scale email corpus training	Presented a compressed transformer architecture that maintains detection accuracy while cutting inference cost for large-scale mail systems.

System Architecture

AI-Based Email Classification and Priority Detection Using Transformers





Engineer to Excel

Module 1: Email Data Collection & Preprocessing



- **Collect Email Data:** Gather emails from public datasets, sample accounts, and organizational email servers.
- **Clean Email Data:** Remove duplicates, errors, irrelevant content, HTML tags, and unwanted characters from the collected emails.
- **Preprocess Text:** Apply tokenization, stop-word removal, and text normalization to create a clean, structured dataset for the Transformer model.

Module 2: Email Classification & Priority Detection Using Transformers

- Use DistilBERT to classify emails into categories such as Personal, Work, Promotional, and Spam.
- Analyze the classified emails to assign High, Medium, or Low priority based on urgency, keywords, sender, and message content.
- Evaluate classification performance using Accuracy, Precision, Recall, and F1-Score, and generate alerts for high-priority emails.



FUTURE ENHANCEMENTS



- Extend the system to support **multiple languages** for email classification.
- Deploy the solution on **cloud platforms** for real-time, scalable usage.
- Integrate with **enterprise email systems** for organization-wide application.
- Implement **continuous learning models** to adapt to evolving email patterns.
- Provide **advanced alerts and cross-platform notifications** for timely responses.



CONCLUSION



- The project successfully implements an **AI-based email classification and priority detection system** using Transformer models.
- Transformer models enable **context-aware understanding** of emails, resulting in **high accuracy** for classification and priority detection.
- The system **reduces manual effort, saves time, and ensures timely response** to important emails.



REFERENCES



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4. Systematic Review of Deep Learning Techniques for Phishing Email Detection,” Electronics, vol. 13, no. 19, 2024.